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Integrating redox-electrodialysis and electrosorption for the removal of ultra-short- to long-chain PFAS

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A major challenge in per- and polyfluoroalkyl substances (PFAS) remediation has been their structural and chemical diversity, ranging from ultra-short to long-chain compounds, which amplifies the operational complexity of water treatment and purification. Here, we present an electrochemical strategy to remove PFAS from ultra-short to long-chain PFAS within a single process. A redox-polymer electrodialysis (redox-polymer ED) system leverages a water-soluble redox polymer with inexpensive nanofiltration membranes, facilitating the treatment of varied chain lengths of PFAS without membrane fouling. Our approach combines both ion migration by electrodialysis (for PFAS with chain lengths ≤C4) and electrosorption strategies (for PFAS with chain lengths ≥C6) to eliminate approximately 90% of ultra-short-, short-chain, and long-chain PFAS. At the same time, we achieve continuous desalination of the source water down to potable water level. The redox-polymer ED exhibits remarkable PFAS removal in real source water scenarios, including from matrices with 10,000 times higher salt concentrations, as well as secondary effluents from wastewaters. Additionally, the removed PFAS is mineralized with a defluorination performance between 76-100% by electrochemical oxidation, highlighting the viability of integrating the separation step with a reactive degradation process.

Per- and polyfluoroalkyl substances (PFAS) are major emerging contaminants to be tackled in clean water production and environmental remediation[1–4]. A key challenge for remediation has been the diversity of PFAS chemical structures, with varied physicochemical properties. To remediate the full range of PFAS contaminants, multiple water treatment processes may be required in sequence. The need for multiple separation units increases operational complexity, capital costs, and energy consumption[5–7]. Recent work has focused mostly on a subset of PFAS (e.g. either long-chain or short-chain)[8–10], with relatively few studies targeting simultaneous removal across the entire range[6,11–13], and no emphasis on tandem desalination with PFAS removal. At the same time, despite the widespread presence of ultra-short-chain PFAS in many drinking water sources and wastewater[14], there is relatively limited research on the removal strategies for these hydrophilic PFAS compounds. Here, we present a single electrochemical process for the removal of ultra-short- to long-chain PFAS, coupled with desalination via the advancement of the redox-mediated electrodialysis system (Fig. 1a). Our approach relies on both ion migration through a reversible redox reaction and electrosorption onto the carbon electrodes to effectively remove PFAS with a wide range of chain lengths. The combination of a redox polymer with cost-effective nanofiltration membranes (NFs) within a redox-mediated electrodialysis platform enables the treatment of ultra-short to long-chain PFAS, while mitigating membrane fouling and improving economic viability. Furthermore, coupling our system with electrochemical oxidation provides a fully electrochemical, process-intensified strategy for the separation and complete mineralization of PFAS.

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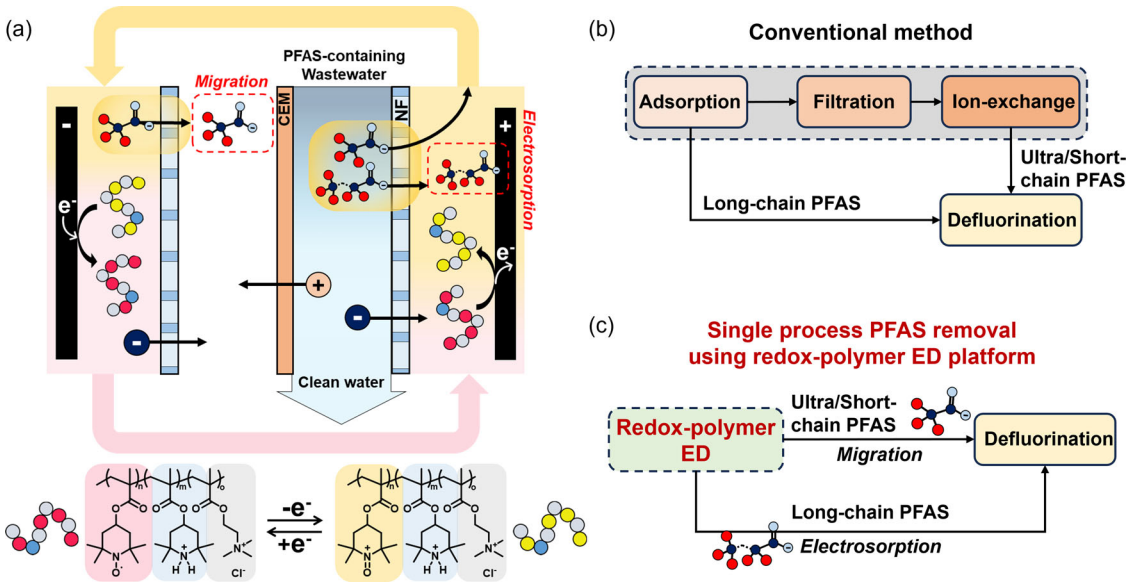


Fig. 1 | Redox-polymer electrodialysis system for PFAS removal. a Schematic illustration of the redox-polymer ED system for PFAS removal (via electrosorption and migration) in tandem with desalination. **b** Conventional strategy to remove ultra-short- to long-chain PFAS with a series of separation processes, including adsorption, filtration, and ion exchange methods. **c** Single process PFAS removal using redox-polymer ED platform, where ultra-short-/short- to long-chain PFAS can be simultaneously removed via migration and electrosorption, respectively. CEM and NF represent cation-exchange membrane and nanofiltration membrane, respectively.

Due to their exceptional chemical stability, thermal resilience, and hydrophobic properties, PFAS are extensively utilized in synthetic chemicals across diverse applications, including coatings, semiconductors, fire retardants, and numerous household products[1,2,14,15]. Since 2009, global regulations have targeted PFAS, starting from long-chain PFAS (C ≥ C8 in carboxylic form; C ≥ C6 in sulfonate form) and their precursors, owing to their potential health effects[15–17]. Consequently, chemical manufacturing has transitioned from long-chain to short-chain PFAS (C3 < C < C8 for carboxylic form). However, recent literature has indicated that shorter-chain PFAS could exhibit persistence and human hazards similar to regulated long-chain PFAS[1,9,14,18]. Moreover, ultra-short-chain PFAS (C ≤ C3 for carboxylic form) are increasingly prevalent in numerous drinking water sources and wastewater, including seawater, rivers, groundwater, and even bottled water (Supplementary Table 1). This occurrence can be attributed to the degradation of longer-chain PFAS and other fluorocarbons into shorter-chain variants[14,15,18–20]. For instance, trifluoroacetic acid (TFA, C2) is a major degradation product of various fluorinated gases used as cooling agents, such as in refrigeration, which then accumulates in the water cycle[21–23]. Consequently, TFA was found in river water (up to 140 ppb, 1.23 μM)[24] and in wastewater treatment plant effluent (206 ppb, 1.81 μM)[25]. Despite the lesser regulatory emphasis on ultra-short-chain PFAS, considering that the drinking water regulation for long-chain PFAS is below 4 ppt[26], these concentrations exceed that limit by up to ten thousand times. Moreover, their smaller sizes, higher mobility, and relatively lower hydrophobicity make them challenging to remove using conventional separation technologies[2,14].

Owing to its unique and diverse physicochemical properties, PFAS cannot be efficiently treated by conventional methods used for the removal of salts or organic substances, including coagulation, aeration, UV oxidation, and chlorination[17,27–29]. In many cases, sequential separation processes are utilized alongside techniques targeting major salts and organic pollutants to effectively eliminate PFAS of varying chain lengths (Fig. 1b). For instance, adsorption with activated carbon has been widely implemented to remove long-chain PFAS, including perfluorooctanoic acid (PFOA, C8)[17,28–30]. However, due to weak hydrophobic interactions between adsorbents and PFAS, most ultra-short- and short-chain PFAS cannot be treated via carbon-based adsorption, especially in the presence of hydrophilic ionic species (e.g., Cl^-, SO_4^{2-}, and NO_3^-)[18,31]. Rather, they are often removed by filtration (e.g., reverse osmosis) and ion exchange (e.g., anion-exchange resins) methods[17,30,32–34]. There has been relatively limited research on the removal of ultra-short-chain PFAS. Recent studies have demonstrated the removal of ultra-short-PFAS via adsorption[35], electrosorption[23], or direct defluorination using photocatalysis techniques[36–38]. Nevertheless, the removal of ultra-short-chain PFAS is challenging in the presence of salts[23,31]. Moreover, due to its low concentration, ultra-short-chain PFAS will eventually require up-concentration to facilitate efficient mineralization.

To address these intrinsic challenges in PFAS removal, we introduce an NF-enabled redox-polymer electrodialysis system (redox-polymer ED) (Fig. 1a). Our redox-polymer ED system replaces expensive and sensitive anion-exchange membranes (AEMs) with affordable size-exclusion NFs for the removal of a wide range of charged species, from inorganic salts to organic substances. In particular, we employed a non-functionalized cellulose-based NF with a 1 kDa molecular-weight cut-off (MWCO). This membrane enables the electric field-driven removal of PFAS without experiencing fouling. We have demonstrated that the combination of water-soluble redox polymer with NFs results in the production of potable water at reduced energy consumption and production cost compared to conventional water purification technologies (e.g., electrodialysis)[39,40]. Building upon our previous research, this study adopts our redox-polymer ED system to treat a wide range of PFAS, from ultra-short- to long-chain PFAS, in tandem with desalination. In particular, we leverage two removal mechanisms for the removal of PFAS: (i) migration for the up-concentration of ultra-short- and short-chain PFAS and (ii) electrosorption on the carbon electrodes for removing long-chain PFAS (Fig. 1a, c), thus demonstrating process intensification of different modes of separation.

The reversible redox reaction of the redox copolymer serves as a driving force for the continuous ion migration of PFAS and salts. Simultaneously, the stationary carbon electrodes facilitate rapid electron transfer to the water-soluble redox polymer, while efficiently electrosorb long-chain PFAS removed from the source water, as demonstrated in prior studies[41,42]. We synthesized a water-soluble redox-active terpolymer, poly(2,2,6,6-tetramethyl-1-piperidinyloxymethacrylate-*co*−2,2,6,6-tetramethyl-1-piperidyl methacrylate-*co*-

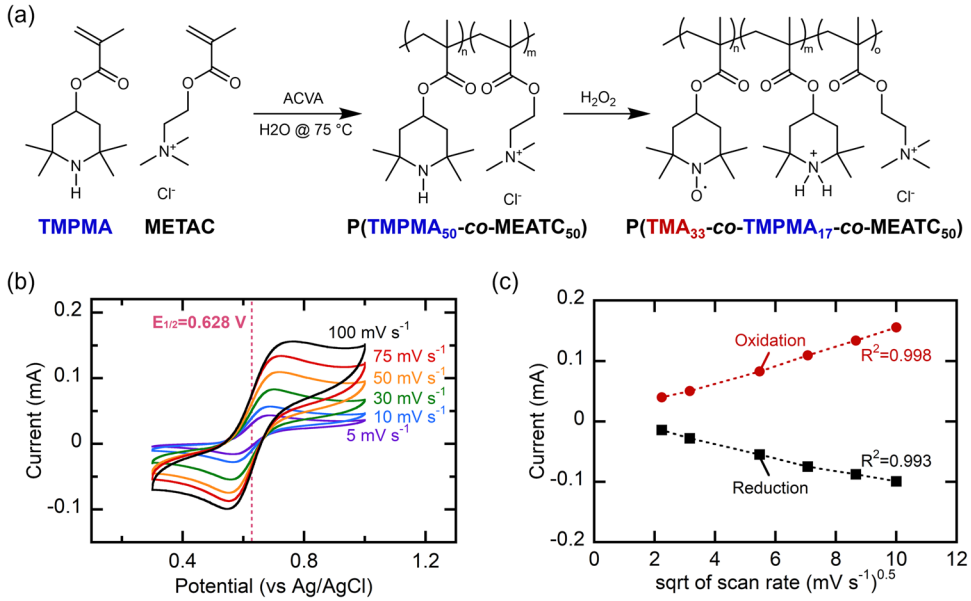


Fig. 2 | Synthesis of redox-active polymer, P(TMA-*co*-TMPMA-*co*-METAC). a. Synthesis of the terpolymer by free-radical copolymerization and chemical oxidation. **b.** Cyclic voltammograms and **c.** Randles-Sevcik plot of P(TMA_{33}-*co*-$TMPMA_{17}$-*co*-$METAC_{50}$) at different scan rates (5, 10, 30, 50, 75, 100 mV/s). ACVA represents 4,4′-Azo-bis-(4-cyanovaleric acid).

[2-(methacryloyloxy)ethyl]trimethyl-ammonium chloride) P(TMA-*co*-TMPMA-*co*-METAC). The redox-active TMA group on the P(TMA-*co*-TMPMA-*co*-METAC) serves as the driving force for desalination and the removal of PFAS from the source water. When the reversible redox species circulates through the anodic and cathodic electrode chambers, it undergoes reversible oxidation and reduction processes at the same $E_{1/2}$, leading to lower energy consumption compared to conventional ED[39,40,43–46]. Additionally, the macromolecular structure of redox-polymer ensures effective retention at the NFs[39,47,48], and controls the direction of desalination and PFAS removal while preventing membrane fouling. Upon the applied voltage, both inorganic anions and PFAS undergo ion migration facilitated by redox reactions and an electric field, crossing over the size-exclusion NF, whereas cations pass through the cation-exchange membrane (CEM) (Fig. 1a). Subsequently, the carbon electrodes within the anodic chamber electroadsorb hydrophobic long-chain PFAS, while ultra-short to short-chain PFAS migrate to the cathodic chamber and cross over the NF to be concentrated in the accumulating channel.

In this study, we demonstrated the unique capability of NF-enabled redox-polymer ED in removing a range of structurally diverse PFAS in dilute concentrations, especially highly challenging ultra-short-chain PFAS (TFA, C2 and perfluoropropanoic acid (PFPrA, C3)), as well as perfluorobutanoic acid (PFBA, C4), perfluorohexanoic acid (PFHxA, C6), and PFOA (C8), in tandem with desalination. Moreover, we assessed the viability of the redox-polymer ED system for the removal of ultra-short-chain PFAS in wastewater treatment, their up-concentration, and electrochemical defluorination. Thus, our redox-polymer ED system demonstrates an energy-efficient and process-intensified platform for PFAS removal through solely electrochemical pathways.

Results and discussion

Design of water-soluble redox-polymer, P(TMA-*co*-TMPMA-*co*-METAC)

The water-soluble redox-active terpolymer, P(TMA-*co*-TMPMA-*co*-METAC) was synthesized through two steps as illustrated in Fig. 2a. P(TMPMA-*co*-METAC) was synthesized through free-radical polymerization involving 2,2,6,6-tetramethyl-1-piperidyl methacrylate (TMPMA) and [2-(methacryloyloxy)ethyl]trimethyl-ammonium chloride (METAC), followed by oxidation of the TMPMA to redox-active 2,2,6,6-tetramethyl-1-piperidinyloxy methacrylate (TMA) using hydrogen peroxide. To prevent potential crossover within NFs (MWCO of 1.0 kDa) during the experiment, we dialyzed the synthesized polymer using NF with an MWCO of 3.5 kDa. Our primary synthesis goal was to maximize the molar ratio of TMA within the terpolymer to enhance overall charge transfer reactions within the terpolymer without using an excessively large amount of the oxidizing agent. Consequently, with the oxidizing agent of 1.5 eq. H_2O_2, over 66% of TMPMA was converted into the redox-active moiety, TMA. The unreacted TMPMA moiety (17 mol% in our terpolymer) enhances the overall solubility of the polymer as it becomes protonated in neutral solution (pK_a of TMPMA: 11.28)[9], as illustrated in Fig. 2a, along with the METAC which provides water solubility to the polymer[49]. For determining terpolymer ratios, we first examined copolymer ratios within P(TMPMA-*co*-METAC) using ^{1}H-NMR (Supplementary Fig. 1a). Subsequently, through UV-vis analysis, we identified the molar ratios of our terpolymer to be P(TMA_{33}-$TMPMA_{17}$-$METAC_{50}$) (Supplementary Fig. 1b) with an average molecular weight of M_n of 2260 g mol^{-1} measured by gel permeation chromatography (GPC) (Supplementary Table 2 and Supplementary Fig. 2).

Cyclic voltammetry (CV) characterization at varying scan rates (5–100 mV s^{-1}) revealed remarkable reversibility of TMA at the $E_{1/2}$ of 0.628 V (Fig. 2b, c). In addition to its rapid electron-transfer properties, P(TMA-*co*-TMPMA-*co*-METAC) exhibits distinct physicochemical and electrochemical properties compared to other water-soluble redox-active polymers, such as ferrocene-based redox-active polymers, enabling effective removal of PFAS in the redox-ED platform. Redox-active units, containing ferrocene- and cobaltocenium, have demonstrated highly effective electrosorption, especially for long-chain PFAS molecules[50]. Consequently, integrating them into our redox-polymer ED platform poses a challenge, as a subsequent desorption step is necessary to release the bound PFAS before the polymer recycling. In this regard, the redox-active moiety, TMA, facilitates redox reaction without direct interactions with PFAS[8–10], enhancing the up-concentration of ultra-short- to short-chain PFAS and promoting the electrosorption of long-chain PFAS on the electrodes.

To further investigate the sorption affinity between redox polymer and PFAS, the binding constant (K_a) between the TMPMA and METAC moieties and PFAS (TFA and PFHxA) was calculated using

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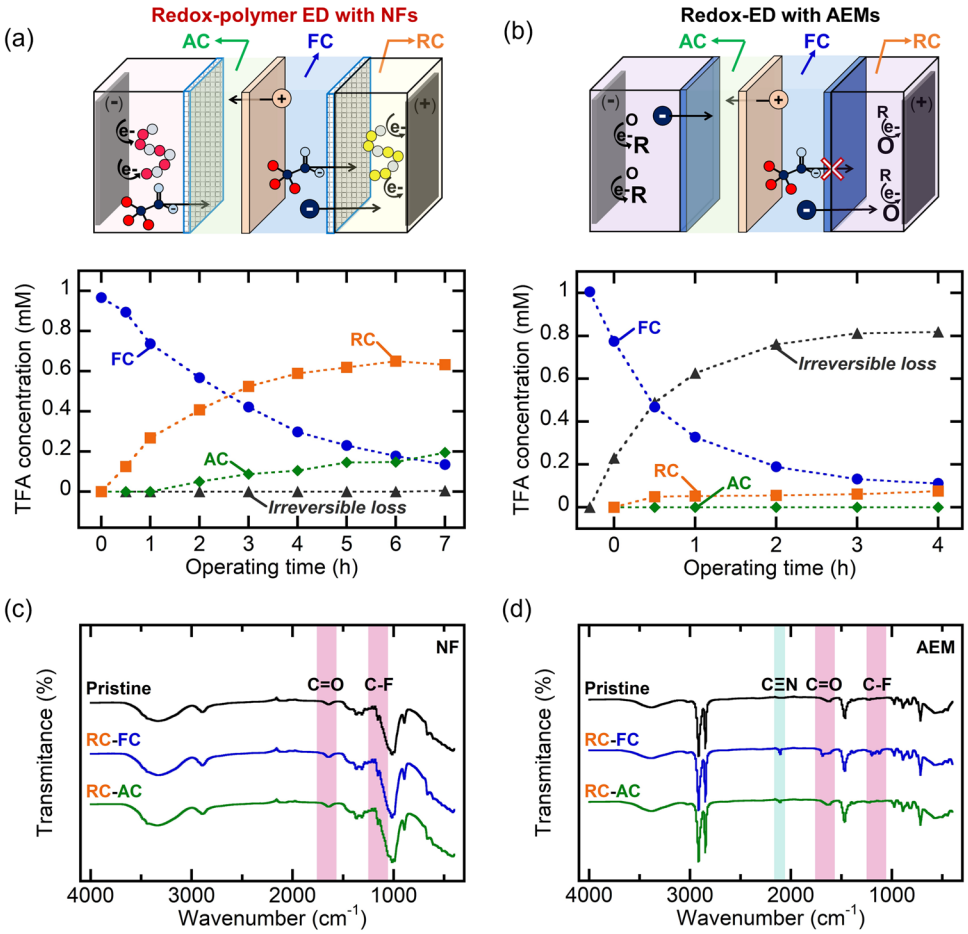


Fig. 3 | Performance of ultra-short-chain PFAS removal. TFA concentration profiles in the feed (FC), accumulating (AC), and redox channels (RC) via **a** the redox-polymer ED system with NFs and **b** the redox-ED system with AEMs. Attenuated total reflectance Fourier transform infrared spectroscopy (ATR-FTIR) spectra of **c** NFs and **d** AEMs after the TFA removal with NF-enabled redox-polymer ED and AEM-enabled redox-ED, respectively. The FC corresponds to the feed channel where the source water circulates, while the AC represents the accumulating channel, where the removed PFAS accumulates. The RC indicates the redox channel, where the redox polymer circulates in cathodic and anodic chambers to facilitate reversible redox reactions.

^{19}F-NMR spectroscopic titrations (Supplementary Figs. 7–10; detailed methodology available in the Methods section titled Polymer binding affinity towards PFAS). The calculated K_a values for TMPMA towards TFA and PFHxA are 9.74 M^{-1} and 40.2 M^{-1}, respectively, and for METAC, they are 8.9 M^{-1} and 38.1 M^{-1}, respectively (Supplementary Fig. 11). Since K_a values for strong binding typically exceed 10^4 M^{-1} [51,52], these results confirm that the TMPMA and METAC moieties in P(TMA-*co*-TMPMA-*co*-METAC), along with TMA, are unlikely to participate in PFAS adsorption. Furthermore, TMA and TMPMA are less hydrophobic than ferrocene[39,53,54], allowing for higher molar ratios of redox-active moieties within the polymer. Consequently, this higher molar ratio of TMA promotes a greater extent of redox reactions at equivalent polymer concentrations compared to other water-soluble polymers. Lastly, both TMPMA and METAC monomers are commercially available at affordable prices ($1 per g_{TMPMA} (Ontario Chemicals, Inc.) and $0.03 per g_{METAC} (Aaron Chem)), facilitating large-scale production.

NF-enabled redox-polymer ED for TFA removal simultaneous with desalination

We performed the removal of the shortest PFAS compound, TFA, using both the redox-polymer ED system with NFs and the redox-ED system with AEMs shown in Fig. 3. To assess its maximum performance of ultra-short-chain PFAS removal, we performed redox-polymer ED with an excess amount of TFA (1 mM) along with 10 mM NaCl in the feed channel (FC). Additionally, we analyzed concentration distributions across the FC, accumulating channel (AC), and redox channel (RC), to investigate any irreversible losses stemming from membrane fouling in both redox-polymer ED with NFs and redox-ED with AEMs. Our NF-enabled redox-polymer ED system exhibited over 86% TFA removal (Fig. 3a), accompanied by an 82% reduction in conductivity (Supplementary Fig. 20a). With a final conductivity of 240 µS, the salt removal has reached levels within the potable water range (<500 µS cm^{-1}), demonstrating effective desalination along with the PFAS removal in a single process. In comparison to the AEM-enabled redox ED system, redox-polymer ED exhibited a 53% enhancement in salt removal kinetics, achieving potable water-level conductivity within 1.8 h (Supplementary Fig. 20). The faster ion removal rate in the redox-polymer ED could be attributed to the physicochemical properties of NF (e.g., large pore size), showing a 54% lower membrane resistance in comparison to the AEM[39], along with its ability to withstand membrane fouling from organic species. Moreover, the TFA concentration profiles across the FC, AC, and RC supported the negligible loss of removed TFA onto the membrane (<3%) due to membrane fouling. The kinetic profiles exhibited that 74% and 23% of the removed TFA were present in the RC and AC, respectively (Fig. 3a).

We conducted a Fourier transform infrared spectroscopy (FTIR) analysis to assess the adsorption of TFA on the NF membrane (Fig. 3c). Given that our NF membrane comprises cellulose, distinct cellulose peaks appear in the pristine FTIR spectrum, notably the C-OH stretching (1015 cm^{-1}), C-O-C stretching of pyranose rings

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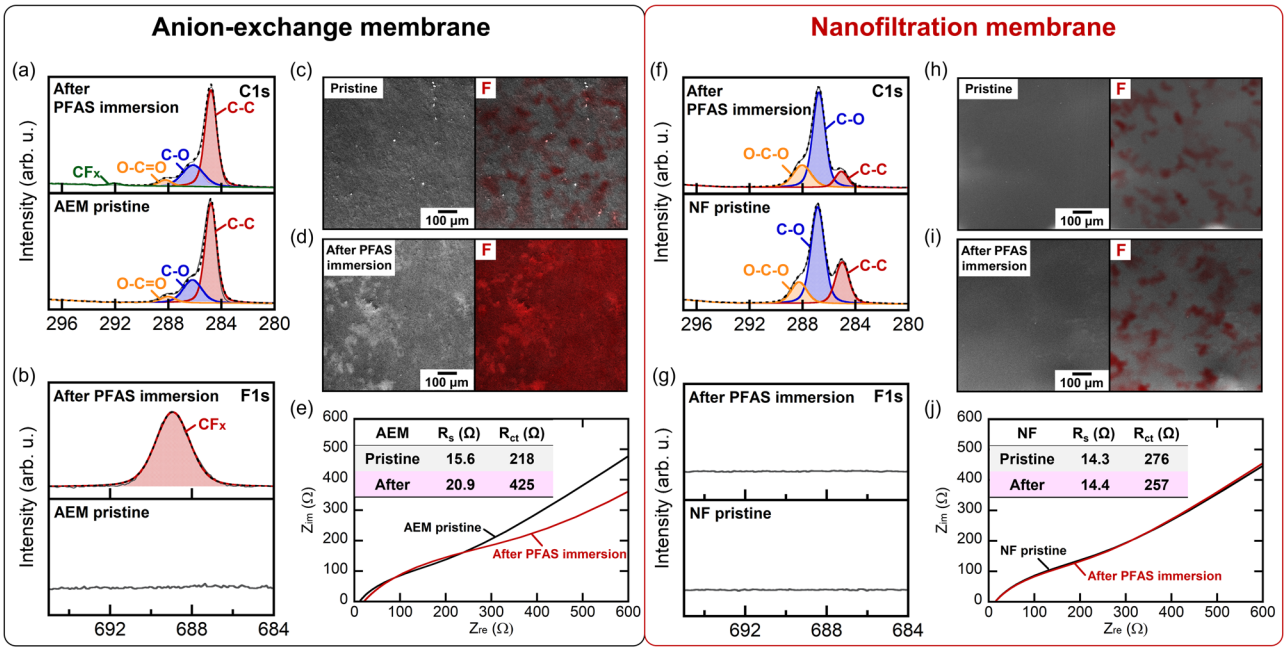


Fig. 4 | Membrane fouling examination. Characterization of AEM: X-ray photoelectron spectroscopy (XPS) spectra of **a** C1s and **b** F1s. Scanning electron microscopy (SEM) images and energy-dispersive X-ray spectroscopy (EDS) mappings (fluoride) for **c** pristine AEM and **d** after fouling test. **e** Nyquist plots of pristine AEM and after PFAS immersion. Characterization of NF: XPS spectra of **f** C1s and **g** F1s, SEM images and EDS mappings (fluoride) for **h** pristine NF and **i.** after fouling test. **j** Nyquist plots of pristine NF and after PFAS immersion. Membranes (2.0 × 2.0 cm^2) were immersed for 24 h in 1 mM TFA, PFPrA, PFBA, PFHxA, and PFOA. R_s and R_{ct} correspond to system and charge-transfer resistances, respectively, and were calculated based on the equivalent circuit of R(Q(RW)). The inset tables in Fig. 4e and j display the values for R_s and R_{ct}.

(1100 cm^{-1}), and O–H stretching (3350 cm^{-1})[55,56]. After conducting 7-hour redox-ED experiments, no significant peak developments related to TFA, such as the C-F peak at 1200 cm^{-1}, were observed, nor were there any notable changes in the C = O peak (Fig. 3c). This spectroscopic analysis aligns with the mass balance calculated in the concentration profiles, emphasizing that the removal of PFAS in the redox-polymer ED platform relies on PFAS migration rather than their irreversible binding to the membranes.

Although the redox-ED with AEMs achieved similar capability in removing TFA (89% TFA removal) and desalination (65% reduction in conductivity), it experienced irreversible adsorption of TFA on AEM, resulting in the loss of over 80% of the initial TFA (Fig. 3b). Notably, more than 23% of TFA was already fouled on the membrane within 10 min before applying voltage, indicating that the membrane fouling primarily stems from the charge interaction between the functional group of the AEM, the quaternary ammonium group, and the carboxylate on the TFA. FTIR analysis of AEMs revealed distinct peaks attributed to C-F and C = O from TFA after the redox-ED experiments, along with a C ≡ N peak from ferri-/ferrocyanide, confirming membrane fouling caused by PFAS and redox species (Fig. 3d). Overall, our NF-enabled redox-polymer ED system effectively addresses the limitation found in AEM-involved systems for ultra-short-chain PFAS removal, while achieving desalination with faster salt removal kinetics.

Investigating membrane fouling by irreversible PFAS adsorption

For a detailed examination of membrane fouling, we performed a membrane adsorption test using 1 mM TFA or the mixture of ultra-short- to long-chain PFAS (1 mM of TFA, PFPrA, PFBA, PFHxA, and PFOA), on AEM, CEM, and NF (Fig. 4 and Supplementary SI 3. Examination of membrane fouling). Each membrane (2.0 × 2.0 cm^2) was immersed in 10 mL of target PFAS solutions containing 10 mM NaCl for 24 h to ensure saturated adsorption of PFAS on membranes (detailed methodologies can be found in the Membrane fouling test section of the Methods). In the presence of 1 mM TFA, more than 80% of the TFA was undetectable in the solution following the immersion of AEM into the TFA solution (Supplementary Fig. 12a). Notable C = O and C-F peak formations on the FTIR spectrum indicate the presence of TFA on the AEM membrane surface (Supplementary Fig. 12b). Moreover, energy-dispersive X-ray spectroscopy (EDS) mappings show a distinct fluoride presence, while scanning electron microscopy (SEM) images reveal increased roughness and non-uniformity compared to pristine AEM, indicating alterations in surface morphology and the formation of aggregates (Supplementary Fig. 13). For NF and CEM, on the other hand, there were negligible changes observed in the concentration of TFA (Supplementary Fig. 12a), the FTIR spectrum (Supplementary Figs. 12c, d), the surface morphology in SEM images, and the fluoride content in EDS mappings (Supplementary Figs. 14, 15). This investigation supports the evidence for successful removal of TFA without any fouling formation on the NF and CEM membranes in the redox-polymer ED system, as seen in Fig. 3a.

The C = O and C-F peak formations on AEM become more distinct in the presence of 1 mM of TFA, PFPrA, PFBA, PFHxA, and PFOA as both concentrations of PFHxA and PFOA decrease by 30% and 80%, respectively (Supplementary Fig. 16a). Furthermore, the X-ray photoelectron spectroscopy (XPS) spectrum reveals significant CFx peaks in both the C1s and F1s spectra following the immersion of AEM in PFAS solutions (Fig. 4a, b). Notable changes in the surface roughness and fluoride content were also observed on SEM images and EDS mappings, respectively (Fig. 4c, d). The irreversible adsorption observed on the AEM is attributed to the presence of the quaternary ammonium functional group, recognized for its strong affinity towards PFAS[57,58]. In fact, this quaternary ammonium group has been demonstrated to be an effective candidate for ion-exchange resin materials for removing PFAS of various chain lengths, including PFBS, PFHxA, and PFOA[11,58–60]. Due to its irreversible PFAS adsorption, AEM suffers performance deterioration, encountering 1.9-fold and 1.3-fold increases in charge-transfer (R_{ct}) and system resistances (R_s), respectively (Fig. 4e).

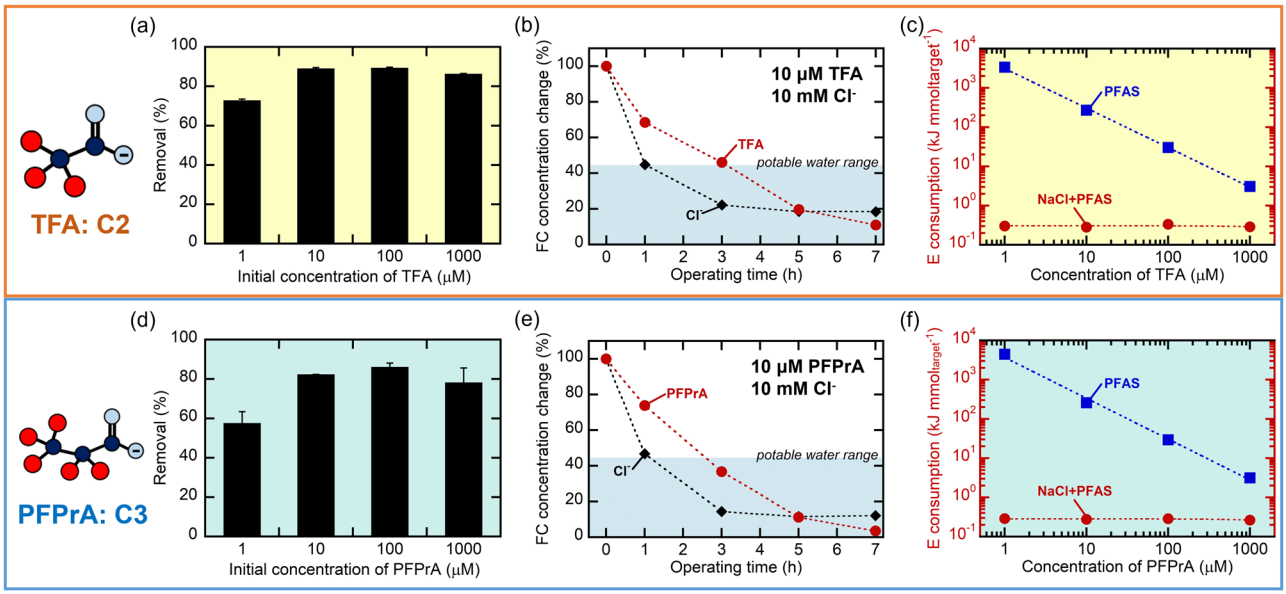


Fig. 5 | Removal of ultra-short-chain PFAS simultaneous with desalination. **a** Total TFA removal %, **b** representative TFA (10 μM) and chloride (10 mM) kinetics in the FC, and **c** energy consumption with respect to the concentration of TFA. **d** Total PFPrA removal %, **e** representative PFPrA (10 μM) and chloride (10 mM) kinetics in the FC, and **f** energy consumption with respect to the concentration of PFPrA. Figure 5a, b were generated with three replicates, while Fig. 5d, e were generated with two replicates. The average values and their standard deviations were plotted.

On the other hand, since both the cellulose-based NF and CEM do not contain cationic groups, they exhibit a low propensity to interact with anionic fluorinated compounds. Through various surface characterization techniques, including XPS, SEM, and EDS, no notable indications of PFAS adsorption were observed for the NF membrane (Fig. 4f–i, Supplementary Figs. 16, 19). Moreover, both charge transfer and membrane resistance remained unchanged after the immersion in the PFAS solution (Fig. 4e), highlighting the remarkable durability of NFs against fluorinated organic compounds. Similar to NF, CEM exhibited no membrane fouling when exposed to a broad range of PFAS, including ultra-short- to long-chain PFAS (Supplementary Figs. 16–18).

Removal of ultra-short-chain PFAS in tandem with desalination

In wastewater or natural water sources, PFAS exists at dilute concentrations (Supplementary Table 1), with salt concentrations thousands of times higher, thus raising the difficulty of PFAS removal[8,9,17,50]. To assess the capability of removing ultra-short-chain PFAS with respect to the range of concentration ratios between PFAS and competing salt, we performed redox-polymer ED at varying concentrations of TFA and PFPrA (from 1 μM to 1 mM), which were 10–10,000 times lower concentrations compared to NaCl (Fig. 5). The concentration ranges of ultra-short-chain PFAS that we investigated are close to those found in real source water, such as wastewater treatment plants (1.8 μM TFA)[25] and river water (1.21 μM TFA)[24] (Supplementary Table 1).

Both TFA and PFPrA showed similar trends in PFAS removal across varying concentrations, demonstrating a rapid increase in removal % from 70% to 89% for TFA and from 57% to 86% for PFPrA as the PFAS concentration increases from 1 μM to 100 μM (Fig. 5a, d). The enhanced removal % is primarily due to the increased availability of PFAS ions compared to chloride (Cl^-) as the concentration ratio between PFAS and Cl^- changes from 1:10,000 (1 μM:10 mM) to 1:100 (100 μM:10 mM). Nevertheless, with the PFAS concentrations increasing from 100 μM to 1 mM, a slight reduction in removal % was observed, resulting in removal % of 86% and 78% for TFA and PFPrA, respectively. Based on the FC concentration changes within 7 h (Supplementary Figs. 21c, 22c), we believe that a longer operating time would facilitate further removal of both TFA and PFPrA at high PFAS concentrations. As noted in Supplementary Table 1, certain source water samples contained ultra-short-chain PFAS exceeding 1.0 μM. Based on our results, our system can treat ultra-short-chain PFAS within a concentration range similar to that observed in actual source water.

We accomplished desalination to potable water standards within 1.5 h with an average Cl^- flux of 33.2 mmol $m^{-2}h^{-1}$ and 38.0 mmol m^{-2} h^{-1} in the presence of TFA and PFPrA, respectively (Supplementary Figs. 21, 22). Interestingly, regardless of the PFAS to Cl^- concentration ratio, salt concentration reached equilibrium within 3 h of operation, achieving an average salt removal of 86% for all PFAS ratios (Fig. 5b, e, Supplementary Figs. 21, 22). On the other hand, throughout the 7-hour experiments, redox-polymer ED continuously removed both TFA and PFPrA, retaining 43% and 58% of the initial TFA and PFPrA removal rate, respectively, by the end of the operation. The remarkable TFA and PFPrA removal, along with the consistent desalination performance, underlines the capability of our NF-enabled redox-polymer ED system for the simultaneous removal of ultra-short-chain PFAS with salt. This result highlights the potential application of our redox-polymer ED systems in tandem treatment of dilute PFAS contaminants with desalination.

Moreover, a significant benefit of simultaneous PFAS removal with desalination using the redox-polymer ED system is the consistent energy consumption of 0.270-0.340 kJ $mmol_{NaCl+PFAS}^{-1}$, regardless of PFAS concentrations as shown in Fig. 5c and f. The mechanism of both PFAS removal and desalination is based on the ion migration to uphold the charge balance of the RC during a reversible redox reaction of the redox polymer. Given that the change in the overall ionic resistance is negligible with respect to the PFAS concentration, both the total energy consumption and overall desalination performance would not be influenced by the concentration of PFAS in the presence of chloride as a predominant species in the FC. Only 0.0075%, 0.11%, 1.1%, and 7.4% of the total energy consumption is allocated to remove 1 μM, 10 μM, 100 μM, and 1 mM ultra-short-chain PFAS, respectively. Consequently, it is more suitable to calculate the overall energy consumption required for both PFAS removal and desalination (kJ $mmol_{NaCl+PFAS}^{-1}$) rather than calculating the energy consumption solely for PFAS removal (kJ $mmol_{PFAS}^{-1}$).

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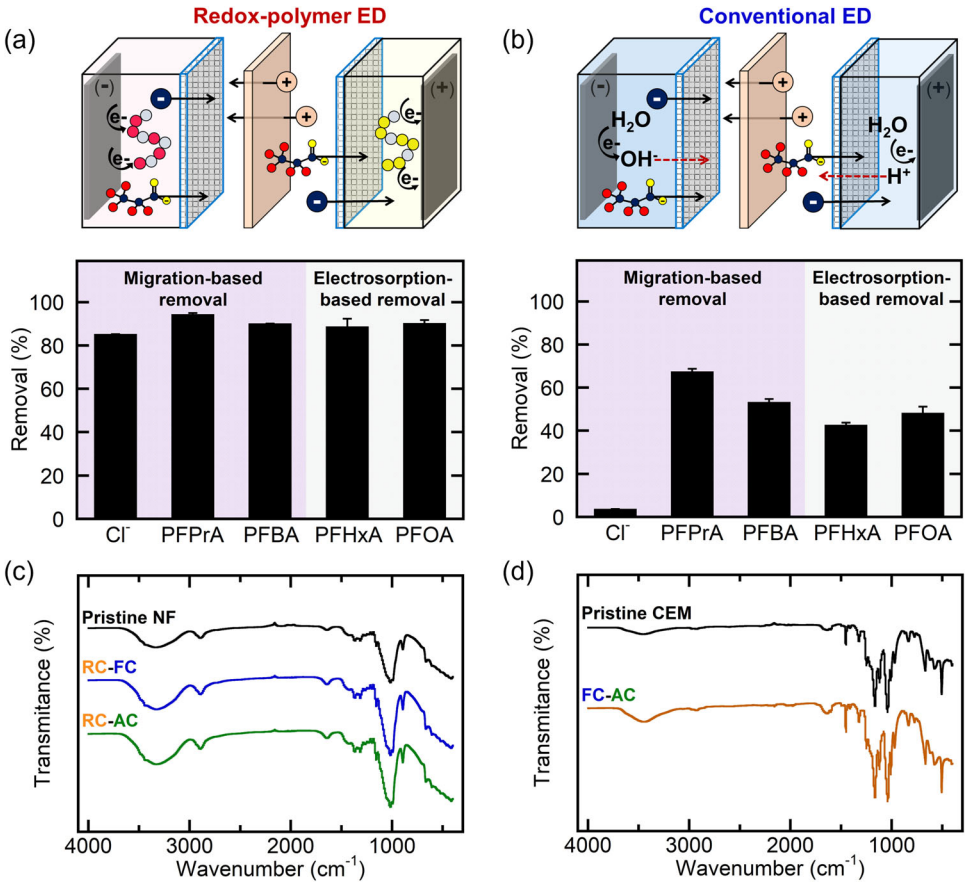


Fig. 6 | Performance of ultra-short-, short-, and long-chain PFAS removal in tandem with desalination. Removal % of 10 mM chloride and 10 μm of PFPrA (C3), PFBA (C4), PFHxA (C6), and PFOA (C8) in the feed channel (FC) via **a** the redox-polymer ED system and **b** conventional ED system with 30 mM NaCl in the RC. FTIR spectra of **c** NFs and **d** CEM after running an experiment with a redox-polymer ED platform. Figure 6a, b were generated with three replicates, and the average values along with their standard deviations were plotted.

Single-process of ultra-short-, short-, and long-chain PFAS removal via redox-polymer ED platform

To demonstrate single-step removal of a wide range of PFAS, we mixed 10 μM of PFPrA, PFBA, PFHxA, and PFOA as representatives of ultra-short-, two short-, and long-chain PFAS, respectively, along with 10 mM NaCl as background electrolyte for desalination. Along with the redox-polymer ED, we performed NF-enabled conventional ED for comparison (Fig. 6). Prior to the comparison, we determined the operating voltage of the conventional ED system through the linear sweep voltammetry, where the current matched that of the oxidation reaction of 30 mM P(TMA_{33}-$TMPMA_{17}$-$METAC_{50}$) at 1.0 V (Supplementary Fig. 23). Our redox-polymer ED system effectively treated a wide range of PFAS, from ultra-short- to long-chain, and exhibited enhanced removal performance in comparison to conventional ED systems using water-splitting reactions for ion removal (Fig. 6a, b). Since both redox-polymer ED and conventional ED used NFs in these studies, a diverse range of PFAS could be removed without any membrane fouling (Fig. 6c, d).

In the redox-polymer ED, we can control the direction of ion removal from the FC to the AC, by taking advantage of the effective retention of the redox-polymer by NFs. Consequently, approximately 90% of PFAS (PFPrA (94%), PFBA (90%), PFHxA (89%), and PFOA (90%)) were removed, regardless of their chain lengths, in addition to an 85% removal of chloride (Fig. 6a). On the other hand, a water-splitting reaction in the conventional ED system generates protons and hydroxides, which can also migrate through NFs in the opposite direction of the intended ion removal. As a result, both salt and PFAS removal performance decreased to 4% and 43–67%, respectively (Fig. 6b and Supplementary Fig. 24). We believe that the limited salt removal in conventional ED with NFs could be attributed to the presence of a higher concentration of ions in the RC than in the FC, leading to a reduction in the diffusion effect and promoting the migration of protons and hydroxides, as observed in conventional ED with AEMs as well[43]. Unlike chloride, the absence of PFAS in the RC facilitated the preferential migration of PFAS over salt removal. This finding suggests that optimizing operating conditions and controlling both diffusion and electrically-driven migration could enhance the rate and selectivity of PFAS removal. Therefore, further studies on diffusion kinetics are necessary to elucidate the correlation between the concentration gradient across channels and the rate and selectivity of ion removal through diffusion.

Furthermore, depending on the PFAS chain lengths, the removal mechanism of PFAS in the redox-polymer ED varied between migration (Supplementary Fig. 25) and electrosorption (Fig. 7e). To probe the underlying removal mechanism, we first conducted electrosorption studies with an equimolar mixture of 1 mM TFA, PFPrA, PFBA, PFHxA, and PFOA in solution along with 10 mM NaCl. The electrosorption experiment was performed in an H-cell using 3.0 × 3.0 cm^2 ACC (with an average weight of 200–250 mg) on both anode and cathode electrodes. The system was operated at a voltage of 1.0 V (two-electrode system) for 1 h to achieve equilibrium electrosorption values. Detailed experimental information can be found in the Methods section titled PFAS electrosorption and defluorination experiments. In the electrosorption test, a significant concentration decrease was observed for PFAS with chain lengths equal to or greater than C6, showing 71% and 88% concentration decreases in PFHxA and PFOA, respectively

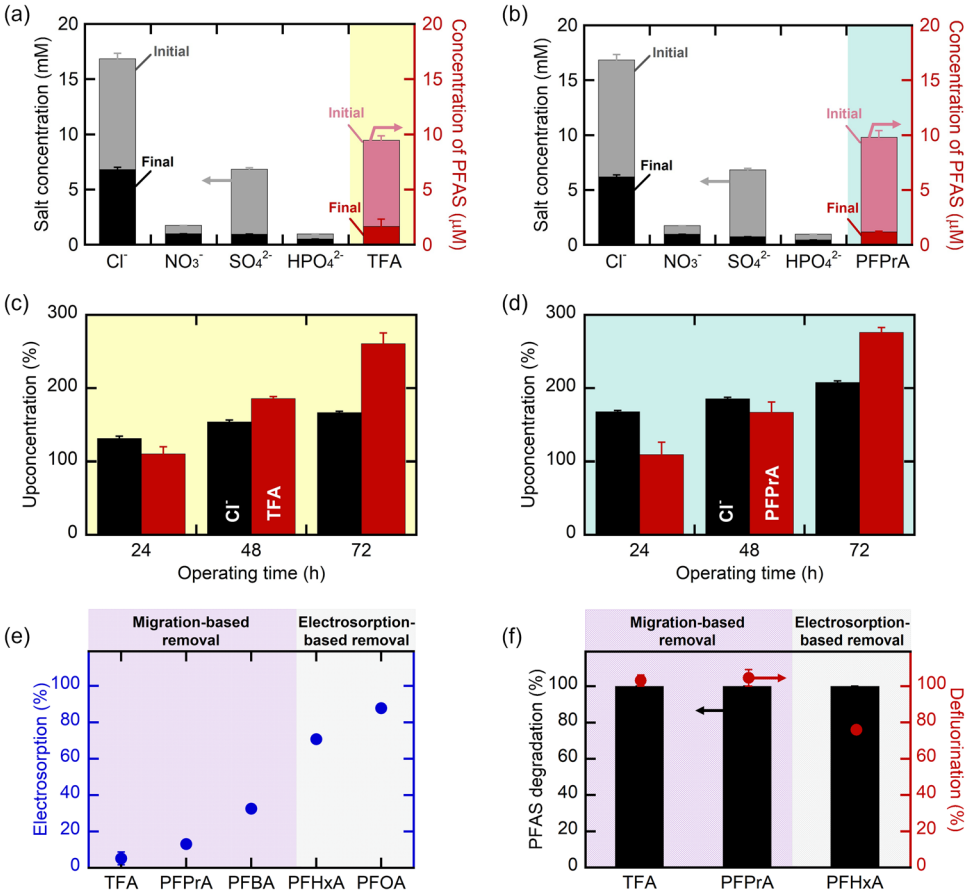


Fig. 7 | Capability of redox-polymer ED for wastewater treatment, up-concentration, and defluorination. Wastewater treatment with **a** TFA (yellow background) and **b** PFPrA (blue background) as representative PFAS contaminants. Up-concentration performance of ultra-short-chain PFAS: **c** TFA and **d** PFPrA. **e** Electrosorption removal performance in a mixture solution of 1 mM TFA, PFPrA, PFBA, PFHxA, and PFOA, demonstrating that PFAS removal with C < 6 is based on the ion migration, while PFAS removal with C ≥ 6 is driven by the electrosorption on the activated-carbon clothes. **f** Defluorination performance of TFA and PFPrA (removal driven by migration) and PFHxA (removal driven by electrosorption). All figure panels were generated with two replicates, and the average values along with their standard deviations were plotted.

(Fig. 7e). Then with a decrease in the chain lengths from C4 to C2, PFAS exhibited weaker binding affinity to the carbon electrodes, leading to a concentration reduction of 33%, 13%, and 5% for PFBA, PFPrA, and TFA, respectively. Our results align with the prior electrosorption studies of PFAS with carbon electrodes and redox-mediated electrodes, showcasing that the electrostatic and hydrophobic interactions effectively eliminate relatively long-chain PFAS, such as PFHxA and PFOA[10,17,41,42,50].

We also assessed the distribution of chloride and PFAS in the redox-polymer ED using different mixtures of PFAS compounds (find a detailed examination in Supplementary Fig. 25). The result highlights that PFAS with chain lengths ≤C4 is predominantly removed through migration from the FC to the AC and RC, while PFAS with chain lengths ≥C6 are treated via electrosorption on the activated-carbon clothes (ACC) electrodes. Overall, by leveraging both ion migration and electrosorption, our redox-polymer ED system can address a wide variety of PFAS compounds in a single operation, effectively overcoming the conventional sequential PFAS treatment process. We believe that the redox-ED platform can intensify the water remediation process, especially advantageous for mixed chain-length PFAS at high concentration levels, as its capacity is not limited by active materials[31], nor does it require high energy inputs such as pressure[17]. We summarized PFAS removal performance across various state-of-the-art technologies in Supplementary Table 4. Note that directly comparing PFAS removal performance across various state-of-the-art technologies can be challenging due to the lack of standardized performance metrics and different initial conditions. Our comparison, based on the removal performance (Supplementary Table 4) and relevant review papers[11,61], offers a preliminary indication that our system is comparable to adsorption, ion exchange, and membrane technologies for long-chain PFAS, while it outperforms adsorption and ion exchange for short-chain PFAS.

Still, to make the system more practical for water treatment and eventual industrial deployment, we need to improve the overall water treatment rate (currently 1.79 L h^{-1} m^{-2}). This can be achieved by increasing the concentration of redox species with stacked electrodes and arranging additional channels (FC and AC) in series, as demonstrated in previous electrodialysis studies[62,63]. Nonetheless, comprehensive evaluations of diverse real-world scenarios—including initial concentrations, specific target PFAS types, and desired target concentrations—are essential. Additionally, rigorous techno-economic analysis must be conducted prior to real-world deployment to ensure the feasibility and effectiveness of our system compared to state-of-the-art PFAS removal methods.

Coupling redox-polymer ED up-concentration with electrochemical defluorination for wastewater treatment

To evaluate the feasibility of our redox-polymer ED system for real wastewater treatment, we performed the desalination on secondary wastewater simultaneously with the removal of ultra-short-chain PFAS (Fig. 7a, b). We introduced 10 μM of TFA or PFPrA to 20 mL of the real wastewater matrix as representative PFAS contaminants (see wastewater matrix in Supplementary Table 5). We performed a real

Table 1 | Removal % of TFA (10 μM) and PFPrA (10 μM) in synthetic brackish water and secondary wastewater

Media	Removal (%)	
	TFA	PFPrA
Synthetic brackish water (10 mM NaCl)	89	85
Secondary wastewater [a]	82	84

[a]Secondary wastewater was collected from the Illinois regional wastewater facility.

wastewater treatment test at 1.0 V for 24 h, using 30 mM of TMA in P(TMA_{33}-*co*-$TMPMA_{17}$-*co*-$METAC_{50}$) as the electrolyte and ACC (3.0 × 3.0 cm^2) on both anode and cathode electrodes. For more detailed experimental procedures, refer to the Investigation of desalination performance and tandem PFAS removal section in the Methods.

In the wastewater, 82% of TFA and 84% of PFPrA were effectively treated (Fig. 7a, b). This removal performance is comparable to treating PFAS in synthetic brackish water (Table 1). Over 65% of small anionic species (61% of Cl^-, 44% of NO_3^-, 88% of SO_4^{2-}, and 50% of HPO_4^{2-}) were desalinated from the wastewater (Fig. 7a, b). In particular, divalent species, such as SO_4^{2-} and HPO_4^{2-}, exhibited more effective desalination than monovalent species, consistent with observations from previous studies[39]. Although multiple electrochemical and pressure-driven studies have demonstrated selective separation of monovalent ions from multivalent species using functionalized NF (e.g., modified polyethersulfone and polyamide) with a MWCO of 200–800 Da[64–67], our approach employs non-functionalized cellulose-based NFs with a larger MWCO of 1 kDa. Consequently, the membrane does not experience significant charge repulsion against multivalent ions. According to the prior study examining the relationship between ion mobility and valency[68], we believe that ion mobility is primarily influenced by the valency of ions under an electric field, possibly due to stronger electrostatic attraction. Additional investigations are needed to correlate ion valency with the mobility within the NF.

Furthermore, we demonstrated the capability of redox-polymer ED to up-concentrate TFA and PFPrA in the AC for energy-efficient mineralization (Fig. 7c, d). In a long-term operation, we utilized recycled redox-polymer to evaluate the reusability and stability of P(TMA_{33}-*co*-$TMPMA_{17}$-*co*-$METAC_{50}$). We first dialyzed the polymers used in previous experiments (primarily for Figs. 5, 6a) to exclusively recover them, and subsequently, we characterized their chemical and electrochemical properties. GPC and CV results confirmed that there was no reduction in polymer chain lengths or changes in electrochemical properties (Supplementary Table 2 and Supplementary Figs. 3, 5), while the FTIR spectrum verified the absence of PFAS in the recycled polymers (Supplementary Fig. 4). The recyclability of polymers will offer additional economic benefits in the redox-polymer ED system, as previously assessed[39], along with the cost-effective polymer NFs and polymer synthesis.

Over 72 h of operation, our system exhibited consistent up-concentration performance for TFA and PFPrA, achieving 2.6- and 2.7-fold up-concentration in the AC compared to the feed solution (Fig. 7c, d). In the initial 24-hour operation, chloride tended to enrich more rapidly than PFAS. However, over an extended period, chloride reached an equilibrium in the AC and exhibited 2-fold enrichment compared to the initial feed solution. This up-concentration experiment provides insights into a single process for PFAS treatment using the redox-polymer ED, which includes desalination and PFAS removal from the feed solution and its up-concentration. Additionally, stable long-term operation highlights the reusability and stability of redox copolymers. However, the polymer recycling process in real-world applications is more complex than in lab-scale settings. Therefore, it is crucial to conduct techno-economic analysis and life-cycle assessments in the future to thoroughly evaluate the feasibility of polymer recycling, considering both economic and environmental implications.

Following redox-polymer ED, treated PFAS can undergo defluorination in two different scenarios: (i) shorter-chain PFAS present in the solution (PFAS with chain lengths ≤C4) and (ii) longer-chain PFAS electrosorbed onto the ACC (PFAS with chain lengths ≥C6). As a proof-of-concept, we demonstrated the defluorination of 1 mM of TFA and PFPrA present in solution (example scenario of (i)) and the defluorination of electrosorbed PFHxA (example scenario of (ii)) (Fig. 7f and Supplementary Fig. 28 for anticipated defluorination process). With boron-doped diamond (BDD) as an anode, TFA and PFPrA showed complete PFAS degradation and defluorination in 6 h, while PFHxA exhibited complete degradation, as well as 76% defluorination in 8 h (Fig. 7f and Supplementary Fig. 26). Electrosorbed PFHxA undergoes two steps of the defluorination process: desorption from the ACC followed by defluorination. The defluorination involves multiple degradation processes, which involve direct electron transfers from PFAS to the BDD surface, and subsequent indirect oxidation of PFAS by hydroxyl radicals generated by the BDD (Supplementary Fig. 28)[69,70]. Consequently, the mineralization of PFHxA requires a longer defluorination period compared to TFA and PFPrA, which are more hydrophilic and do not require a desorption step. Due to high oxidation power, BDD oxidized chloride into perchlorate as a disinfection byproduct[71–75]. Over 80% of chloride (20 mM) present in the defluorination test was converted into perchlorate (Supplementary Fig. 27). Considering the health risks associated with ClO_4^-[71,72,75], reducing chlorine oxidation is essential to improve defluorination efficiency and address a pressing topic in U.S. drinking water safety. In this regard, our system can be integrated with various PFAS degradation systems to address the perchlorate issue, including photoelectrochemical oxidation, plasma-based techniques, and ultrasonic irradiation[17,76–78]. Additionally, surface modification of anode and optimizing oxidation conditions (e.g., adding hydrogen peroxide) can help the electrochemical oxidation process suppress byproduct generation[74,75,78]. Therefore, further investigation into the defluorination process is crucial to envision a sustainable and energy-efficient PFAS treatment process that can be integrated with the redox-polymer ED system for comprehensive PFAS remediation alongside desalination. In an industrial operation, the redox-polymer ED can be upscaled through membrane and electrode stacking and the utilization of a higher concentration of redox-polymers[39,45,62]. After redox-polymer ED operation, the anodic ACC, where the electrosorption of long-chain PFAS occurs, can serve as a cathode for electrochemical defluorination. The AC of the redox-polymer ED system, where ultra-short-/short-chain PFAS is up-concentrated, can serve as an electrolyte for defluorination, effectively integrating two defluorination scenarios into a unified process.

In summary, we integrated a redox-polymer ED platform with an electrosorbent to remove ultra-short- to long-chain PFAS in a single system, by leveraging both ion migration and electrosorption strategies. The combination of affordable and robust NFs with a water-soluble redox-polymer, P(TMA-*co*-TMPMA-*co*-METAC) enables continuous treatment of PFAS across a range of chain-lengths, concurrently with desalination. While conventional redox-ED with AEMs encountered severe membrane fouling when treating ultra-short-chain PFAS, our redox-polymer ED with NFs exhibited 90% removal of ultra-short- to long-chain PFAS without any membrane fouling or polymer crossover. We successfully remediated ultra-short-chain PFAS, even when the salts were present at 10-10,000 times higher concentration than the target PFAS. Simultaneously, the system enabled desalination to potable water salinity levels. Even in comparison to the NF-enabled electrodialysis system, our redox-polymer ED exhibited a 1.7-fold enhancement in PFAS removal and 21.2-fold improvement in desalination performance owing to the remarkable retention of the redox-polymer by NFs.

Moreover, our system enabled the treatment of both TFA and PFPrA from real secondary wastewater, and demonstrated their up-concentration for subsequent electrochemical mineralization. We believe that substituting expensive AEMs with affordable NFs, coupled with the facile polymer synthesis and the reusability of synthesized redox-polymers, can enhance economic feasibility and make the platform applicable to treat a wide range of source waters. Future work could involve the tuning of the molar ratio of P(TMA-*co*-TMPMA-*co*-METAC) for higher removal performance, the exploration of the physicochemical attributes of the nanofiltration membranes regarding PFAS removal efficiency, and the systems-level optimization and integration of the up-concentration step with degradation. Our study presents a proof-of-concept towards process-intensified removal of ultra-short-chain PFAS all the way to long-chain PFAS through a combined membrane- and adsorption-based electrochemical separation.

Methods

Materials and analytic techniques

All chemicals were purchased from Sigma Aldrich, Fisher Scientific, TCI, or VWR, and used as received. Cellulose-based dialysis membranes (Spectra/Por 3) with 1.0 kDa were used as NFs for redox-polymer ED systems, while 3.5 kDa NFs were used for synthesizing redox-polymers. Cation-exchange membranes (CMVN, Selemion) and anion-exchange membranes (AMVN, Selemion) were used ion-exchange membranes for our main system and control experiments. Activated carbon cloths (CH900-20, Kuraray) were used as both cathode and anode.

Chemical and materials characterization

Polymer characterizations. NMR spectra were recorded on a 500 MHz spectrometer with UI500NB to characterize the copolymer ratio between TEMPO and METAC in P(TMPMA-*co*-METAC). NMR sample was prepared with 10 mg of polymer in 500 µL dMeOH solvent. Then we used UV-vis (Cary 60, Agilent Technology) to determine the ratio between TMPMA and TMA, enabling the calculation of overall terpolymer ratios. TEMPO dissolved in MeOH was used as a reference (3–33 µM) for TMA. The synthesized polymers (8–12 mg) were dissolved in MeOH to determine the concentration of TMA in the terpolymer (Supplementary Fig. 1).

Polymer chain lengths were measured by gel permeation chromatography (GPC) using PSS NOVEMA Max (5 mm × 50 mm × 5 µm) and 100 mM NaCl in 0.1 vol% TFA as an eluent. GPC was performed with respect to the poly(2-vinylpyridine) standard with a sample volume of 50 µL at a flow rate of 1 mL min^{-1} (Supplementary Figs. 2 and 3).

Membrane and electrode characterizations. To characterize the membrane physicochemical properties, we monitored attenuated total reflectance Fourier transforms infrared spectroscopy (ATR-FTIR, Thermo Nicolet iS50 FTIR) and X-ray photoelectron spectroscopy (XPS, K-Alpha X-ray Photoelectron Spectrometer, MSE Supplies LLC). For membrane characterizations, we immersed the membranes in DI water overnight and subsequently dried them at 60 °C overnight to eliminate any residual water contents prior to the measurements.

The surface of the activated carbon cloths (ACC) was characterized by scanning electron microscopy (SEM) and energy-dispersive X-ray spectroscopy (EDS) (Axia ChemiSEM, Thermo Fisher) under high vacuum Everhart-Thornley SE detector (ETD) mode with an accelerating voltage of 20 kV and a spot size of 4.5 (Supplementary Fig. 6). Furthermore, the compositions of ACC were analyzed by both CHN (CE440, Exeter Analytical) and halide analyzers (Supplementary Table 3). The ACC was combusted at a high temperature (1800 °C) under He purges. Then the products were mixed with reagents and measured C, H, and N elements by a series of thermal conductivity detectors. The weight percentage of fluoride was also measured by a fluoride selective electrode (Orion Ion Selective Electrodes, Thermo Scientific) following the combustion of solid powders through the Schöniger oxidation method (Supplementary Table 3).

Liquid-phase analytics

The concentration of PFAS was analyzed by high-resolution liquid chromatography-mass spectrometry (LC-MS). For ultra-short-chain PFAS (TFA and PFPrA), Waters Acquity HSS T3 column (2.1 mm × 10 mm × 2.7 µm) was used with the mobile phase of A: 5 mM ammonium acetate and B: acetonitrile at a flow rate of 0.4 mL min^{-1}. An injection volume of 1–10 µL was used with the following elution gradient: 100% phase A for 2 min, 100–0% phase A for 2–2.1 min, 0% phase A for 2.1–4 min, 0–100% phase A for 4–4.1 min, and 100% phase A for 4.1–6 min. Then for short to long-chain PFAS (PFBA, PFHxA, and PFOA), the Acquity BHE C18 column (2.1 mm × 50 mm × 1.7 µm) was used with the same mobile phase and flow rate. An injection volume of 1–10 µL (PFBA) was used with the following elution gradient: 85% phase A for 0.5 min, 85–60% phase A for 0.5–5 min, 60–0% phase A for 5–5.1 min, 0% phase A for 5.1–8 min, 0-85% phase A for 8–8.1 min, and 85% phase A for 8.1–10 min.

The concentration of inorganic anions (Cl^-, NO_3^-, SO_4^{2-}, and HPO_4^{2-}) was measured by Ion chromatography (IC) with the eluent concentration of 30 mM KOH at a flow rate of 1.5 mL min^{-1} with the AS11-HC-4 µm column (4.0 mm × 250 mm). The concentration of ClO_4^- was measured by IC with the eluent concentration of 37 mM KOH at a flow rate of 1.0 mL min^{-1} with the AS25 column (4.0 mm × 250 mm). We diluted samples to ensure that the concentration of each inorganic anion remained below 0.5 mM.

Synthesis of P(TMA-*co*-TMPMA-*co*-METAC)

Prior to the synthesis of the terpolymer, P(TMA-*co*-TMPMA-*co*-METAC), P(TMPMA-*co*-METAC) was synthesized by free-radical copolymerization. 2,2,6,6-Tetramethyl-4-piperidyl methacrylate (TMPMA, 750 mM), [2-(methacryloyloxy)ethyl]trimethyl-ammonium chloride (METAC, 750 mM), and 2-mercaptoethanol (16 mol%) were dissolved in diluted hydrochloric acid solution (the stock HCl was diluted by a factor of 16). The mixture was degassed with argon for 60 min and 4,4'-azo-bis-(4-cyanovaleric acid) (ACVA, 10 mol%) was added before heating the mixture at 75 °C overnight. The solution was then cooled down to room temperature. The sodium hydroxide solution (10 wt%) mixed with hydrogen peroxide (150 mol%), sodium tungstate (1.3 mol%), ethylenediaminetetraacetic acid (EDTA, 0.5 mol%) were added to oxidize TMPMA to 2,2,6,6-tetramethyl-1-piperidinyloxy methacrylate (TMA) and prepare terpolymer, P(TMA-*co*-TMPMA-*co*-METAC). After a day of stirring, additional hydrogen peroxide (150 mol%) was added to the solution and stirred another day. The polymer was dialyzed using a dialysis membrane with a MWCO of 3.5 kDa against water twice and then MeOH once. Next, the polymer was precipitated in acetone, and centrifuged. The centrifuged polymer was dried under vacuum conditions at 60 °C, with a 46% yield of P(TMA-*co*-TMPMA-*co*-METAC).

The ratio between P(TMPMA-*co*-METAC) was calculated by ^{1}H-NMR (500 MHz, dMeOH, δ): 5.0–5.3 (1 H from the tertiary carbon of the piperidyl) and 3.7–3.9 and 4.3–4.7 (2 Hs, methylene group from METAC. Then the final terpolymer ratios were calculated by comprehensive measurements between ^{1}H-NMR and UV-vis (Supplementary Fig. 1).

Polymer binding affinity towards PFAS

To investigate the sorption affinity between redox polymer and PFAS, we performed ^{19}F-NMR spectroscopic titration experiments and calculated the binding constant (K_a). Based on previous studies indicating that TMA does not exhibit a strong binding affinity towards PFAS[10], we focused on the TMPMA and METAC moieties in our terpolymer P(TMA-*co*-TMPMA-*co*-METAC) for major sorption activity. We used a constant concentration of host receptor (20 mM), either TFA or PFHxA, in 0.6 mL of D_2O solution, while varying the guest concentrations of TMPMA and METAC (500 mM stock in D_2O solution) at 0, 0.1, 0.2, 0.3,

0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0, and 2.0 equivalents (Supplementary Figs. 7–10). Assuming 1:1 host-guest binding, we calculated the binding constant using the Benesi-Hildebrand plot as described in the literature[51,52,79]. The linear plot between peak shift^{-1} ($\Delta\delta^{-1}$) and guest concentration^{-1} (C_{moiety}^{-1}), along with the corresponding K_a, is shown in Supplementary Fig. 11.

Electrochemical characterization

To confirm the reversibility of the P(TMA_{33}-$TMPMA_{17}$-$METAC_{50}$), cyclic voltammetry (CV) was performed at varying scan rates of 5, 10, 30, 50, 75, and 100 mV s^{-1}. A three-electrode system with pt wires as working and counter electrodes and Ag/AgCl as a reference was used with a 5 mL solution of 30 mM P(TMA_{33}-$TMPMA_{17}$-$METAC_{50}$) and 100 mM NaCl. CV was IR compensated for the precise Randle-Sevic plotting (Fig. 2c and Supplementary Fig. 5).

Linear sweep voltammetry (LSV) of the redox-polymer ED system was measured to compare the on-set potential of the redox reaction of TMA and oxygen evolution reaction occurring in redox-polymer ED and conventional ED systems, respectively. To disregard the effects of electrosorption on ACC, we ran LSV in two-electrode cells with pt wires as both working and counter electrodes in a batch system from 0 to 1.5 mV s^{-1} at the scan rate of 10 mV s^{-1}. Then the voltage at which the current matched that of the oxidation reaction of P(TMA_{33}-$TMPMA_{17}$-$METAC_{50}$) at 1.0 V was employed as the operational voltage in a conventional ED system for a control experiment (Fig. 6a, b and Supplementary Fig. 23).

Electrochemical impedance spectroscopy (EIS) was performed to analyze the system (R_s) and charge-transfer (R_{ct}) resistances both before and after the PFAS fouling test. The membrane (a circular shape with a diameter of 2 cm) was placed between graphite electrodes (a diameter of 1.8 cm). EIS was performed in a customized coin cell system with 0.5 mL of 10 mM NaCl as an electrolyte. We operated the EIS test in the frequency range of 100 mHz to 500 kHz at 0 V with a voltage amplitude of 10 mV. Subsequently, R_s and R_{ct} were determined by analyzing an equivalent circuit of R(Q(RW)).

Membrane fouling test

The membrane fouling test was conducted using 2.0 × 2.0 cm^2 AEM, CEM, and NF. Prior to the experiments, membranes were immersed in DI water for 2 h to wash any salts from the preservation solution. Then membranes were immersed for 24 h in a 10 mL solution comprising 1 mM of PFAS (single or multiple compounds) and 10 mM NaCl. We collected solution samples both before and after the membrane immersion to quantify the changes in PFAS concentrations. Additionally, for membrane characterizations, such as SEM, EDS, FTIR, and XPS, the immersed membranes were rinsed with deionized water for 1 h and subsequently dried at 60 °C overnight.

Investigation of desalination performance and tandem PFAS removal

The redox-polymer ED system was composed of a redox channel (RC, 4 × 4 × 0.2 cm^3), as well as customized feed (FC, 4 × 4 × 0.3 cm^3) and accumulating channels (AC, 4 × 4 × 0.3 cm^3). In the RC, Ti-mesh was used as current collectors, and ACC (3.0 × 3.0 cm^2 with a weight of 200–250 mg) was used as both cathodic and anodic electrodes. We arranged these channels in the sequence of RC (cathodic chamber)-NF-AC-CEM-FC-NF-RC (anodic chamber) (as illustrated in Fig. 1a). Based on the arrangement sequence, membranes (4.0 × 4.0 cm^2) were positioned between the channels, each having an active surface area of 3.5 × 3.5 cm^2. More detailed configurations and assembly can be found in our previous papers[39,43,44].

For the desalination test, 20 mL of 10 mM NaCl was used in the FC with PFAS (specified in captions for their concentrations and chains) and 20 mL of 10 mM NaCl in the AC. Throughout experiments, 20 mL of 30 mM of TMA in P(TMA_{33}-$TMPMA_{17}$-$METAC_{50}$) was used as an electrolyte in the RC. Unless specified, the system was operated at the flow rate of 10 mL min^{-1} for all FC, AC, and RC within a closed loop. Also, the system was operated in a two-electrode system at the applied voltage of 1.0 V for 7 h. To assess simultaneous desalination performance, the FC conductivity was monitored using a conductivity meter (3574-10 C, Horiba), while IC and LC-MS were employed to measure specific anion compositions and PFAS concentrations, respectively.

For the wastewater desalination test, secondary wastewater from the Illinois regional wastewater facility was used after primary treatment. 20 mL of this wastewater was used as the FC solution with 10 μM of TFA or PFPrA as a representative contaminant. We operated the redox-polymer ED system at the operating voltage of 1.0 V for 24 h.

For up-concentration experiments, 1 L containing 10 mM NaCl with either 10 μM of TFA or PFPrA was used as the FC solution, while 20 mL of 10 mM NaCl and 20 mL of 30 mM of TMA in P(TMA_{33}-$TMPMA_{17}$-$METAC_{50}$) were used as the AC and RC solutions, respectively. The experiments were conducted for three days to investigate long-term stability, as well as the up-concentration capability of the redox-polymer ED system. For up-concentration experiments, recycled redox polymers were utilized to demonstrate the reusability of the polymer materials. After use, redox polymers were recycled using a dialysis membrane with the MWCO of 3.5 kDa against 500 mL of DI water twice and MeOH once. Subsequently, the polymers were recovered by precipitation in acetone. All our experiments were conducted with two to three replicates to ensure system reproducibility, and we plotted the average values along with their standard deviations.

PFAS electrosorption and defluorination experiments

We conducted an electrosorption test in an H-cell to determine the PFAS for which electrosorption is advantageous in their removal. ACC was cut into 3.0 × 3.0 cm^2 for both anode and cathode electrodes. For the anodic electrolyte (working electrode side), a solution of 15 mL containing 1 mM each of TFA, PFPrA, PFBA, PFHxA, and PFOA, along with 10 mM NaCl, was prepared, while the cathodic electrolyte consisted of 20 mM NaCl. We used CEM to partition the anodic and cathodic electrolyte chambers, ensuring that PFAS could neither permeate the membrane nor cause membrane fouling. The H-cell system was operated in a two-electrode system at the operating voltage of 1.0 V for 1 h. The initial and final solutions were collected to analyze PFAS electrosorption using the LC-MS.

To demonstrate the defluorination of ultra-short- to long-chain PFAS, we conducted PFAS oxidation experiments using two distinct approaches. For PFAS treated by up-concentration through ion migration (e.g., TFA and PFPrA), we employed 1 mM of the target PFAS, along with 20 mM NaCl, and conducted the oxidation process at a constant current density of 20 mA cm^{-2}. In our proof-of-concept study, we performed electrochemical oxidation using a BDD anode, known for its exceptional performance in oxidizing organic pollutants[9,10,80,81]. The anode, cathode, and reference electrode were boron-doped diamond (BDD, 1.0 × 1.0 cm^2), ACC (1.0 × 1.0 cm^2), and Ag/AgCl, respectively. In the case of electrosorbed PFAS (e.g., PFHxA), we performed electrosorption of the target PFAS in 1 mM PFAS and 20 mM NaCl prior to the defluorination test at an operating potential of 1.0 V for 1 h in a three-electrode cell with ACC (1.0 × 1.0 cm^2), Pt wire, and Ag/AgCl as anode, cathode, and reference electrode, respectively. Subsequently, we utilized this ACC as a cathode in defluorination experiments to demonstrate the mineralization of electrosorbed PFAS. Fluoride, chloride, and perchlorate concentrations were measured by the IC (refer to the liquid-phase analytics section for specific IC operational parameters). Defluorination (%) was subsequently determined by assessing the mass balance of initial PFAS concentrations in solution (for TFA and PFPrA) or electrosorbed (for PFHxA), whereas the remaining chloride (%) and perchlorate production (%) were calculated based on the starting NaCl concentrations (20 mM).

Data availability

The data supporting the findings of the study are included in the main text and supplementary information files. Source data are provided with this paper. Additional information can be obtained from the corresponding author upon request. Source data are provided with this paper.

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Author contributions

N.K., J.E., and X.S. contributed to the conceptualization. N.K. and E.S. conducted the experiment. N.K. and J.E. were responsible for the synthesis of the materials. N.K. performed the data analysis and wrote the original draft. X.S. supervised the project. N.K., J.E., and X.S. contributed to the editing and reviewing of the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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